



Online service failure and recovery: An integrated meta-analytic perspective of attribution and justice theories

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ABSTRACT

Online service failure is seldom looked into in totality, from causes of failure to recovery outcomes, with reported effects on both its causes and recovery remaining contradictory and inconsistent. To address these issues, this study proposes and empirically validates an integrated *meta*-analytic model, grounded in attribution theory and justice theory, that explains online service failure and the subsequent recovery process, using a cumulative sample of 82,901 individuals from 24 countries across 147 studies. Meta-analytic structural equation modeling (MASEM) indicates that customer-related, service provider-related, and technology-related causes of online service failure increase customer complaints and that effective complaint handling elevates perceptions of distributive, interactional, and procedural justice, each of which strengthens positive emotions that, in turn, raise loyalty, trust, and word-of-mouth while reducing switching. Moderation analysis reveals that the differences in marketing-related factors, including brand prestige (high versus low), offering type (product-based services versus true services versus pseudo services), and purchase frequency (low versus high), together with differences in sample-related factors, such as age (young versus old), gender (male versus female), income (low versus high), and status of country development (developed versus developing), help explain the inconsistencies in reported relationships within online service failure and recovery. Therefore, this study consolidates the evidence on these relationships using the integrated model and, in doing so, advances the literature on online service failure while offering actionable insights for recovery design and execution.

1. Introduction

Online services have grown rapidly as smartphones and social media expand access to digital channels (Azemi et al., 2019). That reach widens firms' customer bases yet raises the likelihood of online service failures, defined as instances where service providers do not meet customer expectations during online transactions (Hazée et al., 2017). Online service failure disrupt service delivery and carry serious consequences, including customer dissatisfaction, negative word-of-mouth, and customer defection, as switching service providers in an online

context requires minimal effort (Fan et al., 2020). In this regard, a clear account of the mechanisms that cause online service failure and how recovery efforts shape subsequent responses is thus necessary. Yet, persistent inconsistencies in reported effects across online service failure call into question the reliability of recovery strategies to reduce negative reactions and produce favorable outcomes.

Previous studies have investigated various facets of online service failure, including its antecedents (Odoom et al., 2020), consequences (Chen & Tussyadiah, 2021), and recovery strategies (Boshoff, 2007), wherein such studies often draw on attribution theory to explain how

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customers assign blame to service providers, distinguishing internal causes, such as service provider shortcomings, from external causes, such as customer or situational factors, and linking these attributions to complaint behavior. Justice theory has also been used to explain how fairness in recovery (i.e., distributive, interactional, and procedural justice) affects customer outcomes (Lim et al., 2025). Yet, findings remain mixed, as some report significant effects for all justice dimensions on satisfaction (Migacz et al., 2018), whereas others find support only for selected dimensions (Nikbin et al., 2015). Similar tensions appear in attribution-based evidence, with some suggesting external attributions predominate (Mezulis et al., 2004) and others pointing to internal factors, such as customer self-efficacy, when explaining causes of failure (Dong et al., 2016). These contradictions, in turn, impede the development of clear guidance for online service failure recovery strategy.

While meta-analytic studies on service failure and recovery exist for offline settings (e.g., Gelbrich & Roschk, 2011; Orsingher et al., 2010; Van Vaerenbergh et al., 2014), online service failure differs in noteworthy ways. For instance, offline service failures often arise from employee behavior, facility breakdowns, inventory shortages, and queue delays, whereas online service failures commonly involve account lockouts, app crashes, checkout errors, cybersecurity breaches, payment failures, system latency, and website usability, among others. Evidence on outcomes diverges as well, wherein loyalty after recovery tends to be higher online (Shankar et al., 2023) while satisfaction and repurchase intention often show stronger effects in offline recovery (Harris et al., 2006). Directly transferring offline insights to online service failure and recovery can, therefore, generate misleading implications.

Accordingly, this study proposes a comprehensive meta-analytic model of online service failure and recovery that integrates attribution theory and justice theory to synthesize fragmented findings and, in turn, provide consolidated estimates spanning the chain from the causes of online service failure to subsequent recovery (Fig. 1 and Table 1). Attribution theory (Weiner, 1985) clarifies who and what are held responsible for online service failure, whereas justice theory (Greenberg, 1990) clarifies how customers judge the fairness of recovery outcomes, procedures, and interactions, and thus their joint use traces the full pathway from failure genesis to recovery evaluation and the behaviors that follow. A meta-analytic approach is appropriate because it cumulates comparable effect sizes and tests moderators across contexts, yielding precise and generalizable estimates from diverse and often times conflicting studies (Lim, 2025). Hence, four questions guide the present study: *first*, what do customers attribute to online service failure; *second*, how do customers react to online service failure; *third*, how do

customers react to strategies that seek to recover from online service failure; and *fourth*, what causes the variations in customer attributions and reactions to online service failure and recovery?

The meta-analysis herein contributes to theory by advancing a unified model that connects two foundational but often siloed perspectives to comprehensively explain the process of online service failure, from causes of failure to recovery outcomes. The synthesis identifies sources of inconsistency across extant studies and clarifies where and when effects are likely to vary, closing a longstanding gap in online service failure research. The analysis also specifies how online service failure differs from its offline equivalent in attribution patterns, recovery expectations, and outcomes. Importantly, the results signal the need to treat online service failure as a distinct phenomenon that warrants strategies designed for the online environment with actionable guidance for tailoring recovery to contexts and segments.

2. Literature review and hypotheses development

2.1. Attribution and outcome of online service failure

Attribution theory (Weiner, 1985), when applied to online service failure, explains how customers assign causes for negative events, distinguishing whether the cause is placed with the customer, the service provider, or technology-related conditions. Noteworthy, prior work often considers service provider-related and technology-related causes as external attributions on the one hand and customer-related causes as internal attributes on the other (Luo & Chea, 2018). Once an online service failure occurs, attribution bias often shifts blame toward the service provider, which increases the likelihood of customer complaints (Bacile et al., 2020). The specific pathways are set out below.

2.1.1. Customer-related factors and customer complaint

Customer-related factors such as low participation in the process (Fan et al., 2020), insufficient knowledge of required steps (Wang & Zhang, 2018), and weak service compatibility (Zolfagharian et al., 2018) can precipitate online service failures. For example, low participation has been linked to higher failure incidence (Vakeel et al., 2018) while limited technology familiarity (Wang & Zhang, 2018) and poor compatibility (Li & Fang, 2016) disrupt service completion. Despite customer-driven origins, complaints frequently target the service provider because of attribution bias (Köcher & Paluch, 2019). Opportunistic complainants may file complaints even when they caused the failure (Reynolds & Harris, 2005), and in a telemedicine context, individuals were more likely to complain regardless of their own role in the failure

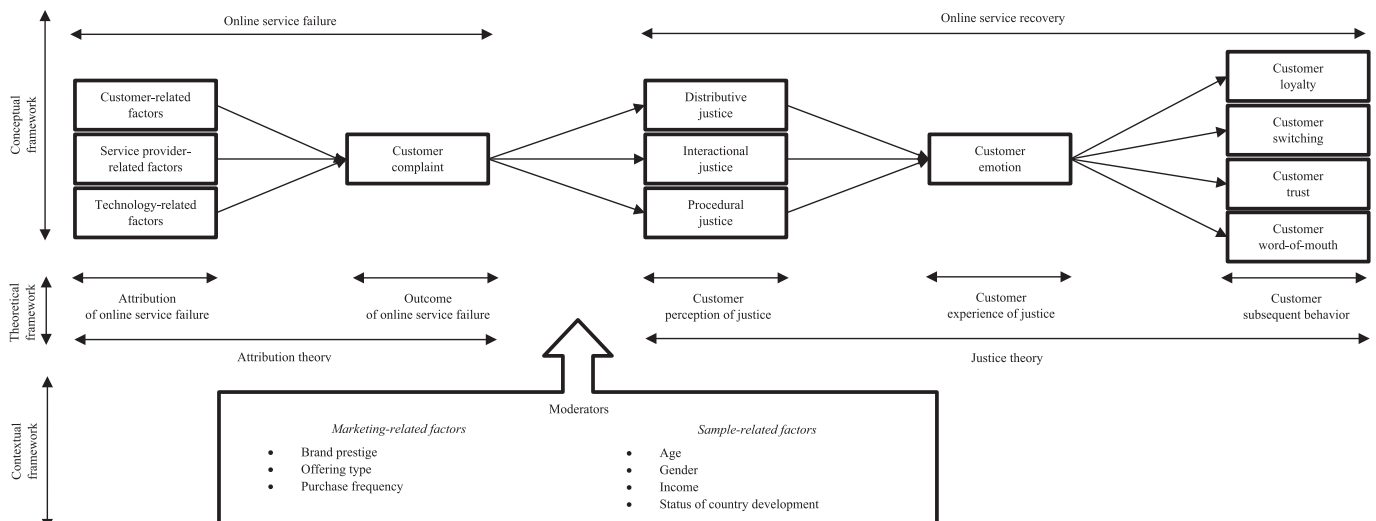


Fig. 1. An integrated meta-analytic model of online service failure and recovery.

Table 1
Concepts in online service failure and recovery.

Concept	Definition	Common aliases	Representative studies
Panel A. Attribution of online service failure			
Customer-related factors	Set of customer attributes responsible for online service failure (Baker & Kim, 2018).	Lack of consumer engagement, Lack of consumer experience online, Lack of e-commerce experience, Lack of fit between online technology and individual value, Lack of participation readiness.	Bagherzadeh et al. (2020), Baker and Kim (2018), Busser and Shulga (2019), Du et al. (2014), Fan et al. (2020), Lin et al. (2018), Ozuem et al. (2024), Sweiss et al. (2022).
Service provider-related factors	Attributes related to the service provider responsible for online service failure (Anwar & Ozuem, 2022).	Informational failure, Lack of commitment of the service provider, Poor e-service quality, Poor employee quality, Poor quality of information.	Anwar and Ozuem (2022), Holloway and Beatty (2003), Jose et al. (2022), Li et al. (2023), Lin et al. (2018).
Technology-related factors	Website-related factors causing difficulty for customers while availing online services leading to potential service failure (Vakeel et al., 2018).	Delayed payment, Incomplete transaction, Poor and complex design, Poor navigation, Website authenticity, Website design complexity.	Goetzinger et al. (2006), Gohary et al. (2016), Köcher and Paluch (2019), Tan et al. (2016), Vakeel et al. (2018), Wang and Zhang (2018).
Panel B. Outcome of online service failure			
Customer complaint	Oral or written communication expressing customers' dissatisfaction (Luo and Chea, 2018)	Complaint intention, Complain tendency, Customer complaining habit, Propensity to complain, Public complaining with social media.	Balaji et al. (2015), Bijmolt et al. (2014), Goetzinger (2007), Luo and Chea (2018), Sengupta et al. (2018), Shams et al. (2020).
Panel C. Customer perception of justice			
Distributive justice	Customers' perceptions of tangible compensations (McColl-Kennedy & Sparks, 2003).	Compensation, Incentive strategy, Non-monetary compensation, Service recovery benefits.	Crisafulli and Singh (2017), Kim and Tang (2016), Mazhar et al. (2022), Tahir (2021), Xu et al. (2020).
Interactional justice	Customers' perception of representatives' apologetic behavior, competence, empathy, honest, and interpersonal sensitivity (Vázquez-Casielles et al., 2012)	Apology, Apology status, Apologetic behavior, Empathy, Forgiveness.	Al-Dmour et al. (2021), Piercy and Archer-Brown (2014), Sengupta et al. (2018), Singh and Crisafulli (2016).
Procedural justice	Customers' perception of complaint-handling process (Ha & Jang, 2009)	Anticipatory justice facets, Informational justice, Procedural remedies, Transparent procedure.	Azemi et al. (2019), Bacile et al. (2020), Weitzl and Hutzinger (2019), Xu et al. (2020).

Table 1 (continued)

Concept	Definition	Common aliases	Representative studies
Panel D. Customer experience of justice			
Customer emotion.	A psychological state eliciting a set of affective responses (e.g., happiness, mental discomfort, regret) post justice assessment (Valentini et al., 2020).	Brand love, Cognitive dissonance, Consumption happiness, Negative affect, Negative emotions, Positive emotions, Regret.	Halpern and Mwesumio (2021), Holloway et al. (2005), Jain and Sharma (2019), Kumari et al. (2021), Robertson et al. (2023), Sharifi et al. (2017), Valentini et al. (2020).
Panel E. Customer subsequent behavior			
Customer loyalty	Customers' willingness to engage with the service provider repeatedly (Gohary et al., 2016).	Attitudinal loyalty, Behavioral loyalty, Repetitive engagement.	Busser and Shulga (2019), Gohary et al. (2016), Kamble and Walvekar (2019), Wu et al. (2020).
Customer switching	Customers' desire to leave the service provider for another (Ozuem et al., 2021).	Perceived betrayal, Psychological contract breach, Switching intention.	Goetzinger (2007), Singh and Rosengren (2020), Tseng (2021).
Customer trust	Customers' belief in the online service provider (Holloway & Beatty, 2003)	Consumer trust, Fulfilment reliability, Perceived trust	Feng and Zhong (2012), Holloway and Beatty (2003).
Customer word-of-mouth	Customers' tendency to share feedback about services post recovery that can be either positive (Azemi et al., 2019) or negative (Odoom et al., 2019).	Likelihood of posting reviews, Motivation to share word-of-mouth, Negative word-of-mouth, Non-favorable word-of-mouth, User motivation for feedback, Word-of-mouth motivation, Word-of-mouth motive	Crisafulli and Singh (2017), Komunda and Osarenkhoe (2012), Luong et al. (2021), Odoom et al. (2019), Sengupta et al. (2024).

(Yan et al., 2021). Some evidence, however, points the other way, whereby customers with high self-efficacy sometimes attempt self-resolution rather than complaining (Chen, 2018). Against this backdrop, the balance of evidence supports a positive link between customer-related causes of service failure and customer complaints. Accordingly:

H₁. Online service failures caused by customer-related factors trigger customer complaints.

2.1.2. Service provider-related factors and customer complaint

Service provider-related triggers include poor service quality (Lin et al., 2018), inaccurate or untimely information (Wang & Zhang, 2018), and weak service provider commitment (Luo & Chea, 2018). Customers often attribute online service failures to the service provider, which raises complaint likelihood (Nugroho & Wang, 2024). For instance, low electronic service quality has been shown to drive complaint behavior in electronic commerce (Rohden and Matos, 2022). Some studies note boundary conditions, whereby strong brand commitment can suppress complaints even when employees cause the failure (Weitzl & Hutzinger, 2019). Despite such contingencies, most findings indicate a positive association between service provider-related causes of service failure and customer complaints. As such:

H₂. Online service failures caused by service provider-related factors trigger customer complaints.

2.1.3. Technology-related factors and customer complaint

Technology-related factors such as website design flaws (Köcher & Paluch, 2019), online payment problems (Gohary et al., 2016), and security concerns (Lin et al., 2018) are prominent causes of online service failure. Given that customers tend to externalize blame in online contexts, such causes are often assigned to the service provider, which increases customer complaint behavior (Fan et al., 2020). Notably, design complexity elevates complaint propensity (Wang & Zhang, 2018) and poor website experiences heighten frustration and complaints (Schiessl et al., 2023). A minority view suggests the bias is not universal, whereby in some self-service technology settings, customers may blame themselves and forgo complaining (Meuter et al., 2000). Even so, the majority of evidence favors a positive relationship between technology-related causes of service failure and customer complaints. As a result:

H₃. Online service failures caused by technology-related factors trigger customer complaints.

2.2. Customer perception of justice

Building on the earlier section, the attribution of causes for an online service failure often leads to complaints, a pattern consistent with attribution bias (Ross, 1977). Grounded in justice theory (Greenberg, 1990), the argument here is that once a complaint is lodged, the manner in which the service provider manages the complaint shapes perceptions of fairness. The literature on online service failure recovery converges on three justice dimensions in the handling of complaints: distributive justice, interactional justice, and procedural justice (Sengupta et al., 2018; Tahir, 2021). While distributive justice concerns the outcomes of recovery, including monetary and non-monetary compensation (McColl-Kennedy & Sparks, 2003), interactional justice concerns the quality of interpersonal treatment during recovery, including empathy, respectful communication, and sincere apology (Vázquez-Casielles et al., 2012), whereas procedural justice concerns the fairness of the recovery process, including transparency, responsiveness, and adherence to clear policies (Ha & Jang, 2009).

Evidence supports the relevance of each dimension, though not without debate. Tangible recovery outcomes anchor perceived distributive justice and often matter to customers who bear direct loss, with several studies emphasizing their centrality (Lim et al., 2025) while others argue that outcomes work best when paired with fair procedures and respectful interactions (Odoom et al., 2020). Empathetic communication and credible apologies tend to improve perceived interactional justice (Che et al., 2022), yet some studies caution that interpersonal treatment, on its own, may be insufficient to shift overall fairness judgments (Urueña & Hidalgo, 2016). Transparent and responsive procedures generally bolster perceived procedural justice (Che et al., 2022), though a few findings question its influence in certain settings (Wu & Huang, 2015). In this regard, the weight of evidence indicates that when complaint handling demonstrates fair outcomes, fair interpersonal treatment, and fair procedures, customers report higher post-recovery perceptions across all three justice dimensions. Consequently:

H₄. Handling customer complaints in ways that exhibit distributive, procedural, and interactional justice enhances post-recovery perceptions of (a) distributive, (b) procedural, and (c) interactional justice.

2.3. Customer experience of justice

While the previous section established how complaint handling shapes perceptions of distributive, procedural, and interactional justice, the next link in the present section concerns customer emotion. Perceived justice, understood as the alignment between what customers expect and what they receive in outcomes, interpersonal treatment, and processes, shapes emotional responses in online service failure (Köcher & Paluch, 2019). When received justice meets or exceeds expectations, customers report positive emotions such as relief and happiness, whereas when received justice falls short, customers report negative

emotions such as anger, discomfort, disappointment, dissonance, and regret (Bahri Ammari et al., 2022). Findings, however, are not uniform across studies, as some report that distributive and interactional justice strengthen positive emotion while procedural justice does not (Kuo & Wu, 2012), whereas others find that procedural justice exerts the primary influence with limited effects from distributive and interactional justice (del Río-Lanza et al., 2009). Even with such variation, the weight of evidence supports a positive link from each justice dimension to emotion, which motivates the following hypotheses:

H₅. Customer experience of (a) distributive, (b) interactional, and (c) procedural justice have a positive impact on their emotion.

2.4. Customer subsequent behavior

Emotions that arise from justice evaluations are linked to key behavioral outcomes, including loyalty, switching, trust, and word-of-mouth (Azemi et al., 2019; Gohary et al., 2016; Manu et al., 2021; Ozuem et al., 2017). However, evidence on how emotions translate into these outcomes is not uniform, even when those emotions stem from distributive, procedural, and interactional justice. For instance, emotions, both positive and negative, have been shown to influence loyalty after evaluations of interactional and distributive justice, with limited effects observed after procedural justice (Chebat & Slusarczyk, 2005). Likewise, more recent evidence shows that positive emotions following distributive justice can stimulate favorable electronic word-of-mouth, whereas negative emotions do not always trigger unfavorable electronic word-of-mouth (Fu et al., 2015). Others show that when interactional and procedural justice are perceived as high in responses to online complaints, trust tends to rise through positive emotions, whereas the converse holds when negative emotions are salient (Olson & Ro, 2020). Similarly, lower perceived justice can intensify negative emotions and raise switching intentions while higher perceived justice can temper such intentions (Mody et al., 2020). Against this backdrop, the weight of evidence indicates that emotions formed in response to perceived recovery fairness shape downstream behavior. Given this:

H₆. Customer emotion after experiencing distributive, interactional, and procedural justice impacts their loyalty, switching, trust, and word-of-mouth.

2.5. Boundary conditions

The proposed meta-analytic model integrates attribution theory and justice theory. Even so, estimated relationships among model variables vary across extant evidence on online service failure. For example, perceptions of distributive and interactional justice have been found to shape emotions and intention to engage in word-of-mouth (Jung & Seock, 2017), whereas other work reports effects only for distributive justice, with limited or no effects for interactional or procedural justice (Choi & Choi, 2014). Meta-analysts commonly attribute such inconsistencies to differences in contextual and methodological features across studies (Hazée & Van Vaerenbergh, 2021). The present review identifies three *contextual*, marketing-related factors—i.e., brand prestige (high versus low), offering type (product-based services versus true services versus pseudo services), and purchase frequency (high versus low)—and four *methodological*, sample-related factors—i.e., age (young versus old), gender (male versus female), income (high versus low), and status of country development (developed versus developing)—as likely moderators.

2.5.1. Marketing-related factors

Brand prestige¹ (high versus low). Brand prestige reflects the perceived status and value customers ascribe to a brand (Tse et al.,

¹ A brand was categorized as low or high prestigious brand using the information available in each study.

2023). Higher prestige can raise expectations for recovery outcomes, interpersonal treatment, and procedures, whereby unmet expectations may depress perceived justice and weaken positive emotion and trust (Lu et al., 2020). Countervailing evidence suggests that prestige can signal credibility and reduce uncertainty, thereby strengthening positive emotions and trust when recovery is handled well (Joe et al., 2017). Variation in expectations and credibility cues provides a clear rationale for moderation. In this regard:

H₇ Brand prestige (high vs. low) moderates the relationships in the meta-analytic model.

Offering type (product-based services versus true services versus pseudo services). Offering type shapes both attribution and recovery expectations. Product-based services involve tangible goods transacted through online platforms, such as books sold on Amazon (Gu & Tayi, 2017). True services are intangible and consumed at delivery, such as streaming or cloud computing (Shobeiri et al., 2015). Pseudo services blend elements of both, such as online group buying that bundles products with service-like features (Gu & Tayi, 2017). Evidence indicates that customers may be less inclined to blame providers in close, relational true-service contexts (Tsai et al., 2014), whereas blame attribution tends to be higher for product-based services when failures occur (Wan et al., 2011). Differences in tangibility, relational intensity, and process visibility therefore offer a strong basis for moderation. Hence:

H₈ Offering type (product-based services versus true services versus pseudo services) moderates the relationships in the meta-analytic model.

Purchase frequency² (high versus low). Purchase frequency captures how often customers use an online service (Chang et al., 2012). Infrequent purchasers have been shown to spread more negative word-of-mouth after dissatisfying failures than frequent purchasers (Holloway et al., 2005). Justice components may also matter differently by frequency segment, for instance, interactional treatment tends to lift emotions and satisfaction among low-frequency customers, whereas procedural clarity and consistency tend to matter more for high-frequency customers (Chang et al., 2012). These patterns imply heterogeneous paths across frequency groups. On this basis:

H₉ Purchase frequency (high versus low) moderates the relationships in the meta-analytic model.

2.5.2. Sample-related factors

Age (young versus old). Age differences are likely to shape paths in the meta-analytic model. In online gaming, distributive, procedural, and informational justice show overall positive effects on satisfaction and subsequent electronic word-of-mouth, yet younger gamers share more electronic word-of-mouth than older gamers, in part because peer guidance in gamer communities amplifies sharing among younger users (Ding & Lii, 2016). In group service failure settings, younger customers have been found to be more active in spreading negative word-of-mouth, consistent with stronger negative emotion and greater weight placed on perceived causes of failure (Albrecht et al., 2017). These patterns point to age-contingent differences in how justice evaluations translate into emotion and behavior. Accordingly:

H₁₀ Age (young versus old) moderates the relationships in the meta-analytic model.

Gender (male versus female). Gender-based variation also appears in service failure research. Some studies report that women exhibit greater loyalty to service providers when recovery is handled well (Morgeson et al., 2020) while others find that men report higher loyalty under effective recovery (Azemi et al., 2019). In mobile shopping, men's higher perceived ease of use of technology has been associated with stronger attribution of responsibility to providers for technology-related

failures and with a higher propensity to complain (Assaker, 2020). Such differences suggest gendered pathways from attribution and justice to emotion and downstream behavior. As such:

H₁₁ Gender (male versus female) moderates the relationships in the meta-analytic model.

Income (high versus low). Income can shift which justice components matter most. Among higher-income customers, distributive justice has been shown to exert the strongest effect on emotion and positive word-of-mouth after recovery, followed by procedural justice, whereas among lower-income customers, procedural justice takes precedence, followed by distributive justice (Keaveney & Parthasarathy, 2001). Arguably, higher-income customers can better anticipate likely compensation and therefore focus on outcomes, whereas lower-income customers face greater ambiguity about entitlements and therefore weigh process fairness more heavily. Therefore:

H₁₂ Income (high versus low) moderates the relationships in the meta-analytic model.

Status of country development (developed versus developing). The development status of a country may also condition effects. In developed countries, service providers tend to be blamed when they cause a failure but not when customers do (Weber & Sparks, 2010), whereas in developing countries, service providers are often blamed regardless of who caused the failure (Matikiti et al., 2019). Expectations differ as well, whereby customers in developed countries tend to hold higher service quality expectations and, consequently, demand higher compensation during recovery than customers in developing countries (Malhotra et al., 2005). Such differences in blame attribution and expectation levels provide a clear basis for moderation. Thus:

H₁₃ Status of country development (developed versus developing) moderates the relationships in the meta-analytic model.

3. Methodology

3.1. Corpus curation

3.1.1. Search

An article search was conducted using the keyword “online service failure,” together with the alternatives “service failure and e-retailing” and “service failure and online booking,” across major databases and publisher platforms, including Academia.edu, EBSCO, Emerald, Gale, Google Scholar, J-Gate, JSTOR, ResearchGate, Sage, ScienceDirect, Science Open, Scopus, Springer, Taylor & Francis, and Web of Science. The same keywords were used to locate relevant theses on ProQuest and unpublished studies on the Social Science Research Network (SSRN). Consistent with prior meta-analyses (e.g., Mehta et al., 2022), a call for unpublished studies was issued via the Electronic List for Marketing Academics and Researchers (ELMAR). Reference lists of online service failure reviews were also screened to identify additional articles. This search process, in line with meta-analytic practice (Jebarajakirthy et al., 2024), yielded 301 articles.

3.1.2. Filter

Following the removal of 123 duplicates, the remaining 178 documents were retained if they (a) were written in English, (b) reported scholarly research, (c) examined online service failure—service failures occurring in online purchasing or online interactions with a firm or other stakeholders, (d) used quantitative methods, and (e) reported zero-order correlations or sufficient statistics for conversion (e.g., Student's *t*). Applying the criteria led to the exclusion of 22 non-English articles, eight non-scholarly articles, three articles not focused on online service failure, and nine non-quantitative articles, resulting in 136 articles. Four articles that did not report the necessary (correlation) statistics were subsequently removed, resulting in 132 articles. At revision, 15 additional relevant articles were identified and added, bringing the final total to 147 (Fig. 2).

² To determine customers' purchase frequency, we searched for the information regarding the number of times a customer purchased a specific product/service examined in each study.

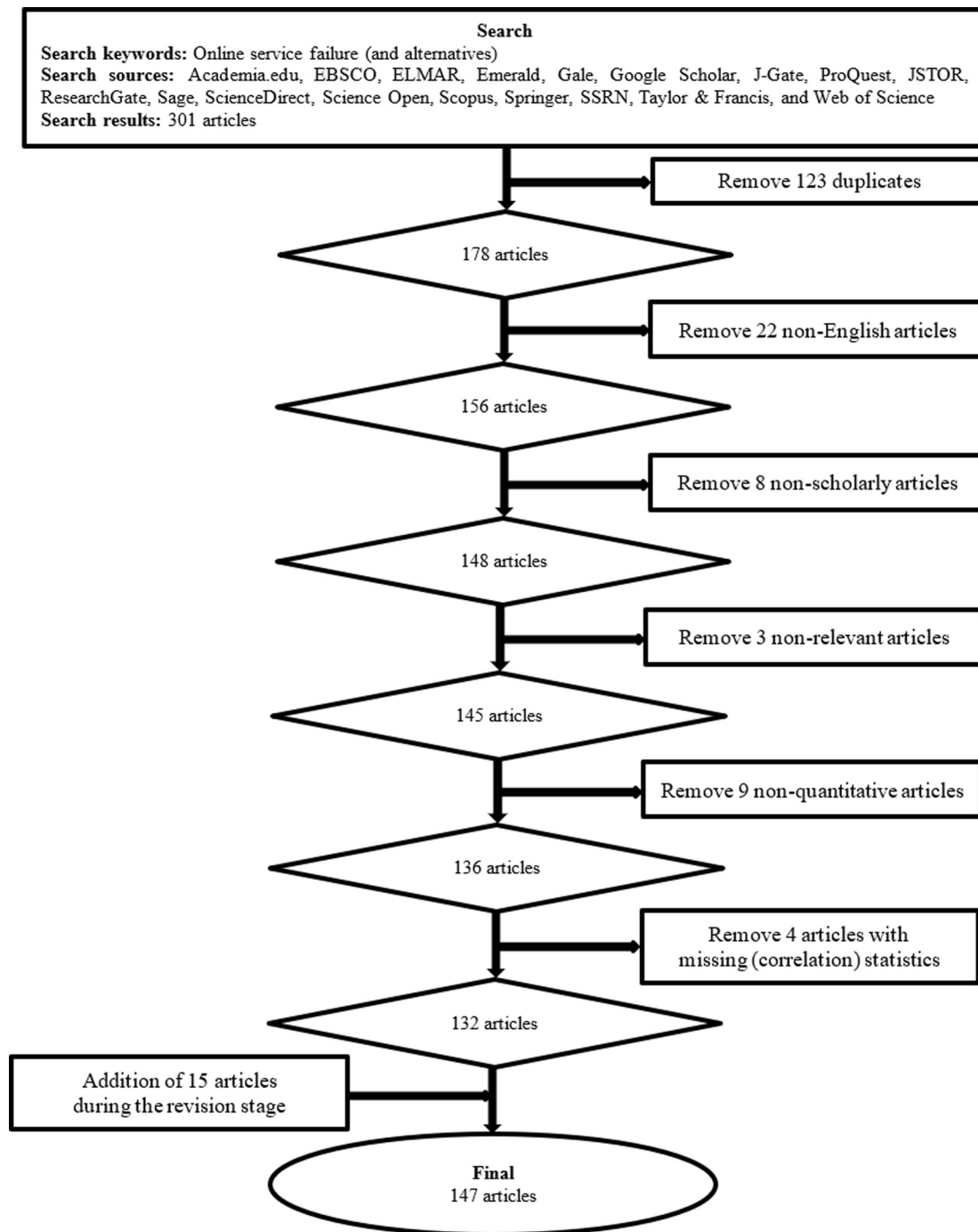


Fig. 2. Search and filter process.

3.1.3. Coding

The 147 articles cover a cumulative sample of 82,901 individuals across 24 countries. Several constructs shared definitions but appeared under different labels. For example, “compensation” (Tahir, 2021) and “service recovery benefits” (Mazhar et al., 2022) describe similar content but use different terms. Such constructs were coded according to their stated operationalization. Two researchers conducted the coding. Krippendorff’s α indicated 93 % agreement, supporting the reliability of the coding process (Maseeh et al., 2021).

3.2. Meta-analysis

3.2.1. Assessment of publication bias

Publication bias was assessed using funnel plots (Maseeh et al., 2021), Rosenthal’s fail-safe N (Rosenthal, 1979), and Egger’s regression (Mehta et al., 2022). In funnel plots, symmetry indicates a lower risk of publication bias (Hunter et al., 2014), whereas fail-safe N was computed to estimate the number of missing studies that would be required to nullify the observed effects, wherein values exceeding five times the number of included studies (k) plus 10 ($5k + 10$) suggest limited influence of publication bias (Lipsey & Wilson, 2001) while a non-significant intercept for Egger’s regression indicates no systematic small-study effects (Mehta et al., 2022).

3.2.2. Bivariate analysis

Statistical power for each bivariate relationship was examined to gauge the likelihood of Type II error, that is, failing to reject a false null hypothesis (Mehta et al., 2022). Analyses used zero-order correlations. When zero-order correlations were unavailable, other statistics (e.g., F ratios) were converted to correlations (Ashaduzzaman et al., 2022). Mean observed correlations and reliability-corrected correlations were then estimated. A random-effects model corrected correlation coefficients for measurement and sampling error by dividing each raw correlation by the square root of the associated Cronbach's α (Chang & Taylor, 2016). After reliability correction, correlations were transformed to Fisher's z (Mehta et al., 2022), weighted by the inverse of their variances, and converted back to correlations for interpretation (Ashaduzzaman et al., 2022). Confidence intervals at the 95 % level were used to judge significance, whereby estimates whose intervals excluded zero were treated as significant at $p < 0.05$ (Borenstein et al., 2011).

3.2.3. Meta-analytic structural equation modeling (MASEM)

Meta-analytic structural equation modeling (MASEM) was conducted in SPSS AMOS v.27 to test direct paths. A pooled, meta-analytic correlation matrix served as the input. Variable means were set to zero and standard deviations to one.

3.2.4. Assessment of heterogeneity

Heterogeneity across studies was examined using the Q statistic, the 75 % rule of thumb, and 80 % credibility intervals (Mehta et al., 2022). A significant Q suggests that sampling error alone cannot account for observed variance, implying the presence of moderators. Likewise, under the 75 % rule, if less than 75 % of observed variance is attributable to sampling error, effect sizes are considered heterogeneous and potential moderators are implicated. Similarly, wide 80 % credibility intervals point to meaningful between-study variation consistent with moderator influence.

3.2.5. Moderation analysis

Meta-regression was used to estimate the effects of candidate moderators on relationships within the model (Zablah et al., 2012). Several primary studies reported multiple effect sizes, which create dependence among estimates (Maseeh et al., 2021). A three-level mixed-effects specification addressed this structure (Van Vaerenbergh et al., 2014), wherein level one captured sampling variance for each effect size, level two captured within-study variance across multiple effect sizes from the same study, and level three captured between-study variance reflecting true differences across studies (Khamitov et al., 2024). Moderator definitions and coding appear in Table 2. Regression estimates were computed for the influence of each moderator on each relationship in the meta-analytic model.

3.2.6. Multicollinearity diagnostics

Potential multicollinearity was evaluated in three steps. First, pairwise correlations were inspected, where no zero-order correlation exceeded 0.80, indicating limited risk (Mehta et al., 2022). Second, variance inflation factor (VIF) estimates were below five, consistent with the absence of problematic multicollinearity (Hair et al., 2009). Third, sensitivity analyses sequentially excluded constructs involved in pairwise correlations greater than 0.50 (Eisend & Tarrahi, 2022), with results for remaining constructs unchanged, thus reinforcing the conclusion that multicollinearity does not affect the results.

4. Results

4.1. Results of publication bias tests

Publication bias was assessed using funnel plots (Hunter et al., 2014; Maseeh et al., 2021), Egger's regression (Mehta et al., 2022), and

Table 2

Operationalization and coding scheme of moderators.

Moderator	Operationalization	Coding
Panel A. Contextual: Marketing-related moderators		
Brand prestige	Whether the perceived status associated with the brand is low or high.	1 = High (97) 2 = Low (50)
Offering type	Whether the offering was product-based services, true services, or pseudo services.	1 = Product-based services (67) 2 = True services (46) 3 = Pseudo services (34)
Purchase frequency	Whether the purchase frequency is low or high	1 = Low (94) 2 = High (53)
Panel B. Methodological: Sample-related moderators		
Age	Average age of a study's sample.	Continuous variable
Gender	Proportion of females (versus males) in the sample of a study.	Continuous variable
Income	Whether the average income of respondents in the sample was low or high in relation to the median income in their country.	1 = Low (59) 2 = High (88)
Status of country development	Whether the data were collected in developed or developing country	1 = Developed (91) 2 = Developing (56)

Rosenthal's fail-safe N (Rosenthal, 1979). The funnel plots were symmetric, indicating little risk of bias. As shown in Table 1, fail-safe N values exceeded the $5k + 10$ benchmark for the relevant numbers of studies, with a mean fail-safe N of approximately 399, implying that about 399 null-effect studies would be required to overturn the observed relationships. Egger's regression intercepts were non-significant for the hypothesized paths, consistent with the absence of small-study effects (Table 3).

4.2. Results of bivariate analysis

Statistical power for each bivariate relationship exceeded 0.50 (Table 3), indicating a low risk of Type II error (Ellis, 2010). Random-effects, reliability-corrected correlations show that customer-related factors ($\rho = 0.236^{***}$), service provider-related factors ($\rho = 0.261^{***}$), and technology-related factors ($\rho = 0.317^{***}$) are each positively associated with customer complaints. Customer complaints, in turn, are positively associated with distributive justice ($\rho = 0.255^{***}$), interactional justice ($\rho = 0.315^{***}$), and procedural justice ($\rho = 0.240^{***}$). Distributive justice ($\rho = 0.260^{***}$), interactional justice ($\rho = 0.262^{***}$), and procedural justice ($\rho = 0.190^{***}$) are positively associated with customer emotion, which is positively related to customer loyalty ($\rho = 0.340^{***}$), customer trust ($\rho = 0.334^{***}$), and customer word-of-mouth ($\rho = 0.318^{***}$) and negatively related to customer switching ($\rho = -0.312^{***}$).

4.3. Results of MASEM

The meta-analytic model fit the data well: Relative chi-square (χ^2/df) = 2.435 ($p < 0.05$), Comparative fit index (CFI) = 0.958, Goodness of fit index (GFI) = 0.963, Root mean square error of approximation (RMSEA) = 0.082, Standardized root mean square residual (SRMR) = 0.060. Fig. 3 and Table 4 report path estimates.

Customer-related factors ($\beta = 0.350^{***}$), service provider-related factors ($\beta = 0.319^{**}$), and technology-related factors ($\beta = 0.344^{***}$) had significant positive effects on customer complaints, explaining 32 % of its variance ($R^2 = 0.32$), thus supporting H_1 , H_2 , and H_3 .

Customer complaints were positively associated with distributive justice ($\beta = 0.299^{**}$), interactional justice ($\beta = 0.323^{**}$), and procedural justice ($\beta = 0.270^*$), explaining 20 % of variance for distributive justice ($R^2 = 0.20$), 22 % for interactional justice ($R^2 = 0.22$), and 19 % for procedural justice ($R^2 = 0.19$), thus supporting H_{4a} , H_{4b} , and H_{4c} .

Table 3

Results of bivariate analysis, publication bias tests, and heterogeneity analysis.

Bivariate relationship	Bivariate analysis						Publication bias tests		Heterogeneity analysis					
	N studies	N samples	r ⁺	ρ	95 % confidence interval		Statistical power	Egger's test	N _{FS}	Q statistic	df (Q)	75 % rule	80 % credibility intervals	
					Lower	Upper							Lower	Upper
Panel A. Attribution of online service failure														
Customer-related factors → Customer complaint	19	8435	0.222	0.236***	0.151	0.308	> 0.999	0.53	312	1589.30***	18	36 %	−0.647	0.850
Service provider-related factors → Customer complaint	21	10,237	0.254	0.261***	0.102	0.328	> 0.999	0.17	342	1574.32***	20	15 %	−0.524	0.875
Technology-related factors → Customer complaint	18	8746	0.302	0.317***	0.213	0.421	> 0.999	0.15	298	1422.21***	17	20 %	−0.215	0.924
Panel B. Outcome of online service failure and customer perception of justice														
Customer complaint → Distributive justice	20	8587	0.254	0.255***	0.180	0.357	> 0.999	0.34	406	1622.71***	19	18 %	−0.295	0.762
Customer complaint → Interactional justice	22	9573	0.326	0.315***	0.223	0.450	> 0.999	0.28	493	1892.47***	21	30 %	−0.556	0.906
Customer complaint → Procedural justice	16	7466	0.235	0.240***	0.192	0.308	> 0.999	0.19	352	1436.56***	15	24 %	−0.771	0.863
Panel C. Customer experience of justice														
Distributive justice → Customer emotion	12	3873	0.256	0.260***	0.157	0.415	> 0.999	0.28	209	256.98***	11	20 %	−0.637	0.778
Interactional justice → Customer emotion	17	6056	0.261	0.262***	0.115	0.396	> 0.999	0.24	179	140.56***	16	22 %	−0.134	0.752
Procedural justice → Customer emotion	15	4421	0.184	0.190***	0.042	0.298	> 0.999	0.17	211	153.44***	14	13 %	−0.028	0.963
Panel D. Customer subsequent behavior														
Customer emotion → Customer loyalty	35	12,853	0.325	0.340***	0.208	0.556	> 0.999	0.40	762	1950.10***	34	12 %	−0.471	0.868
Customer emotion → Customer switching	21	8079	−0.300	−0.312***	−0.487	−0.251	> 0.999	0.20	639	702.89***	20	36 %	−0.202	0.865
Customer emotion → Customer trust	32	12,046	0.316	0.334***	0.147	0.501	> 0.999	0.36	658	1875.36***	31	29 %	−0.386	0.759
Customer emotion → Customer word-of-mouth	24	9873	0.302	0.318***	0.241	0.469	> 0.999	0.27	332	589.32***	23	27 %	−0.506	0.877

Notes: N = Cumulative studies (N studies) or sample size across the studies (N samples) reporting a particular bivariate relationship. r^+ = Simple average correlation. ρ = Correlation corrected for measurement and sampling error. N_{FS} = Fail-safe N . df = Degrees of freedom. *** = $p < 0.001$.

Distributive justice ($\beta = 0.280^{**}$), interactional justice ($\beta = 0.300^{**}$), and procedural justice ($\beta = 0.204^*$) had significant positive effects on customer emotion, explaining 31 % of its variance ($R^2 = 0.31$), thus supporting H_{5a} , H_{5b} , and H_{5c} .

Customer emotion was positively associated with customer loyalty

($\beta = 0.384^{***}$), customer trust ($\beta = 0.368^{***}$), and customer word-of-mouth ($\beta = 0.353^{***}$) and negatively associated with customer switching ($\beta = -0.341^{***}$), explaining 26 % of variance for customer loyalty ($R^2 = 0.26$), 17 % for customer switching ($R^2 = 0.17$), 29 % for customer trust ($R^2 = 0.29$), and 25 % for customer word-of-mouth ($R^2 =$

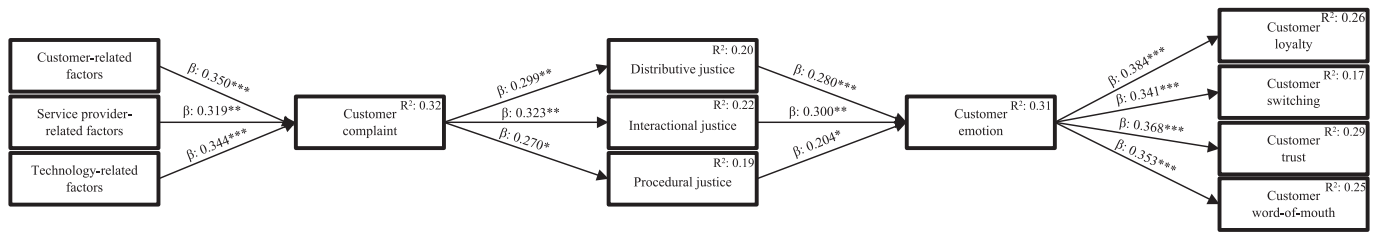


Fig. 3. Meta-analytic structural equation modeling (MASEM) of online service failure and recovery. Notes: * = $p < 0.05$. ** $p < 0.01$. *** = $p < 0.001$. β = Regression coefficient. R^2 = Coefficient of determination. Relative chi-square (χ^2/df) = 2.435 ($p < 0.05$). Comparative fit index (CFI) = 0.958. Goodness of fit index (GFI) = 0.963. Root mean square error of approximation (RMSEA) = 0.082. Standardized root mean square residual (SRMR) = 0.060.

Table 4
Results of meta-analytic structural equation modeling (MASEM).

Bivariate relationship	Hypothesis	Estimate
Panel A. Attribution of online service failure		
Customer-related factors → Customer complaint	H ₁	0.350***
Service provider-related factors → Customer complaint	H ₂	0.319**
Technology-related factors → Customer complaint	H ₃	0.344***
Panel B. Outcome of online service failure and customer perception of justice		
Customer complaint → Distributive justice	H _{4a}	0.299**
Customer complaint → Interactional justice	H _{4b}	0.323**
Customer complaint → Procedural justice	H _{4c}	0.270*
Panel C. Customer experience of justice		
Distributive justice → Customer emotion	H _{5a}	0.280**
Interactional justice → Customer emotion	H _{5b}	0.300**
Procedural justice → Customer emotion	H _{5c}	0.204*
Panel D. Customer subsequent behavior		
Customer emotion → Customer loyalty	H _{6a}	0.384***
Customer emotion → Customer switching	H _{6b}	−0.341***
Customer emotion → Customer trust	H _{6c}	0.368***
Customer emotion → Customer word-of-mouth	H _{6d}	0.353***

Notes: * = $p < 0.05$. ** = $p < 0.01$. *** = $p < 0.001$. Relative chi-square (χ^2/df) = 2.435 ($p < 0.05$). Comparative fit index (CFI) = 0.958. Goodness of fit index (GFI) = 0.963. Root mean square error of approximation (RMSEA) = 0.082. Standardized root mean square residual (SRMR) = 0.060.

0.25), thus supporting H_{6a}, H_{6b}, H_{6c}, and H_{6d}.

4.4. Results of heterogeneity analysis

Heterogeneity analysis points to meaningful variation across all bivariate links. All Q statistics were significant (***, $p < 0.001$), with values ranging from 140.56 to 1950.10 and degrees of freedom from 11 to 34 (Table 3). The 75 % percent rule values fell between 12 % and 36 %, that is, well below the 75 % threshold, indicating that sampling error cannot account for the observed variance. The 80 % credibility intervals were wide and crossed zero in every case (lower bounds from − 0.771 to − 0.028 and upper bounds from 0.752 to 0.963), consistent with true heterogeneity and the presence of moderators. These results support the examination of the moderators specified earlier.

4.5. Results of moderation analysis

The meta-regression estimates for moderators are illustrated in Fig. 4 and reported in Table 5.

Regarding brand prestige, significant moderation appears on multiple paths, including the effect of service provider-related factors ($\beta = 0.102^*$) and technology-related factors ($\beta = 0.322^*$) on customer complaint, the effect of customer complaint on distributive justice ($\beta = 0.363^*$), interactional justice ($\beta = 0.350^*$), and procedural justice ($\beta = 0.426^*$), the effect of interactional justice on customer emotion ($\beta =$

0.458***), and the effect of customer emotion on customer loyalty ($\beta = 0.393^*$), customer switching ($\beta = - 0.407^*$), customer trust ($\beta = 0.380^*$), and customer word-of-mouth ($\beta = 0.425^*$), thus supporting H₇.

Regarding offering type, significant moderation is evident for the effect of customer-related factors ($\beta = 0.127^*$) and service provider-related factors ($\beta = 0.351^*$) on customer complaint, the effect of interactional justice on customer emotion ($\beta = 0.113^*$), and the effect of customer emotion on customer trust ($\beta = 0.124^*$), thus supporting H₈.

Regarding purchase frequency, significant moderation occurs for the effect of customer-related factors on customer complaint ($\beta = 0.103^*$), the effect of distributive justice on customer emotion ($\beta = 0.434^*$), and the effect of customer emotion on customer loyalty ($\beta = 0.356^*$), customer switching ($\beta = - 0.389^*$), and customer word-of-mouth ($\beta = 0.481^*$), thus supporting H₉.

Regarding age, significant moderation appears for the effect of customer-related factors on customer complaint ($\beta = 0.121^*$), the effect of customer complaint on distributive justice ($\beta = 0.404^*$) and interactional justice ($\beta = 0.105^*$), and the effect of customer emotion on customer loyalty ($\beta = 0.427^*$), thus supporting H₁₀.

Regarding gender, significant moderation is observed for the effect of customer-related factors ($\beta = 0.463^*$), service provider-related factors ($\beta = 0.455^*$), and technology-related factors ($\beta = 0.309^*$) on customer complaints, the effect of interactional justice on customer emotion ($\beta = 0.397^*$), and the effect of customer emotion on customer loyalty ($\beta = 0.112^*$) and customer switching ($\beta = - 0.105^*$), thus supporting H₁₁.

Regarding income, significant moderation is present for the effect of procedural justice on customer emotion ($\beta = 0.395^*$) and the effect of customer emotion on customer loyalty ($\beta = 0.474^*$) and customer word-of-mouth ($\beta = 0.386^*$), supporting H₁₂.

Regarding country development status, significant moderation holds for the effect of service provider-related factors ($\beta = 0.100^*$) and technology-related factors ($\beta = 0.416^*$) on customer complaint, the effect of interactional justice on customer emotion ($\beta = 0.124^*$), and the effect of customer emotion on customer trust ($\beta = 0.103^*$), thus supporting H₁₃.

5. Discussion

Prior meta-analyses centered on offline service failure (e.g., Gelbrich & Roschk, 2011; Orsingher et al., 2010; Van Vaerenbergh et al., 2014) do not fully capture the dynamics of online settings, where a global customer base, always-on channels, and instantaneous amplification via social platforms heighten the salience of failure and recovery (Bart et al., 2005). This study synthesizes 147 articles spanning 82,901 participants across 24 countries within an integrated model grounded in attribution theory and justice theory. The evidence traces a coherent chain: online service failures, attributed to customer-related, service provider-related, and technology-related causes, elevate customer complaint; effective handling of customer complaint raises perceived distributive, procedural, and interactional justice; perceived justice shapes customer emotion; and customer emotion, in turn, influence their loyalty, switching, trust, and word-of-mouth. Effect magnitudes vary across

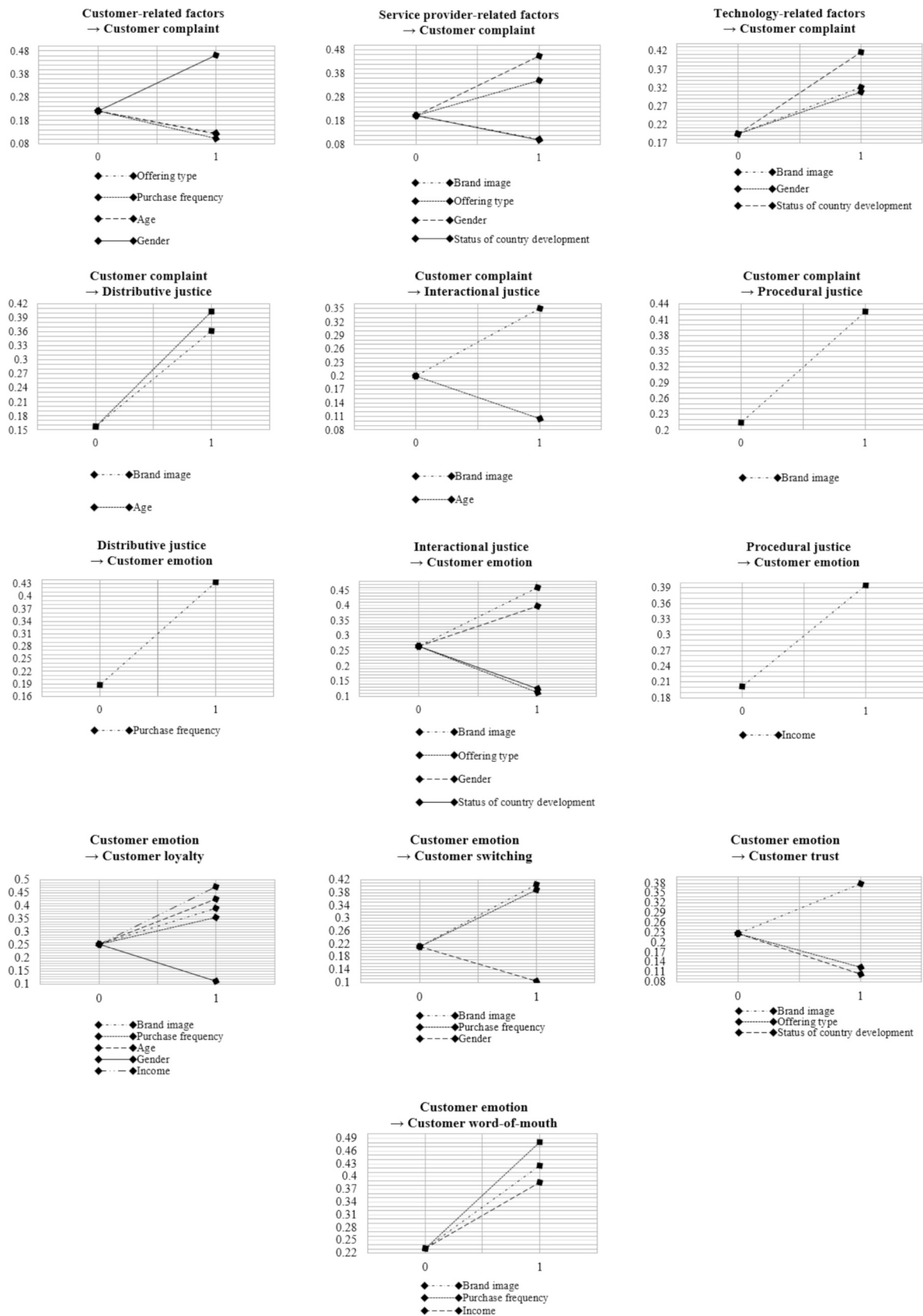


Fig. 4. Results of moderation analysis. Notes: 0 represents the intercept, 1 represents the regression coefficient for the second condition of each moderator.

Table 5
Results of moderation analysis.

Bivariate relationship	Intercept	Brand prestige (H ₇)	Offering type (H ₈)	Purchase frequency (H ₉)	Age (H ₁₀)	Gender (H ₁₁)	Income (H ₁₂)	Status of country development (H ₁₃)
Panel A. Attribution of online service failure								
Customer-related factors → Customer complaint	0.221	—	0.127**	0.103*	0.121*	0.463***	—	—
Service provider-related factors → Customer complaint	0.203	0.102*	0.351**	—	—	0.455***	—	0.100*
Technology-related factors → Customer complaint	0.195	0.322**	—	—	—	0.309*	—	0.416***
Panel B. Outcome of online service failure and customer perception of justice								
Customer complaint → Distributive justice	0.158	0.363**	—	—	0.404***	—	—	—
Customer complaint → Interactional justice	0.200	0.350**	—	—	0.105**	—	—	—
Customer complaint → Procedural justice	0.214	0.426***	—	—	—	—	—	—
Panel C. Customer experience of justice								
Distributive justice → Customer emotion	0.188	—	—	0.434***	—	—	—	—
Interactional justice → Customer emotion	0.264	0.458***	0.113*	—	—	0.397**	—	0.124*
Procedural justice → Customer emotion	0.202	—	—	—	—	—	0.395**	—
Panel D. Customer subsequent behavior								
Customer emotion → Customer loyalty	0.254	0.393**	—	0.356**	0.427***	0.112*	0.474***	—
Customer emotion → Customer switching	−0.212	−0.407**	—	−0.389**	—	−0.105*	—	—
Customer emotion → Customer trust	0.227	0.380**	0.124**	—	—	—	—	0.103*
Customer emotion → Customer word-of-mouth	0.231	0.425***	—	0.481***	—	—	0.386**	—

Notes: * = $p < 0.05$. ** = $p < 0.01$. *** = $p < 0.001$. “—” = Results are not available.

studies as a function of marketing-related factors (e.g., brand prestige, offering type, purchase frequency) and sample-related factors (e.g., age, gender, income, status of country development). The subsections that follow discuss main (direct) effects first, then moderation (conditional) effects.

5.1. On online service failure and recovery

5.1.1. From causes of failure to customer complaint

Online service failures attributed to customer-related, service provider-related, and technology-related factors each increase customer complaint, with the strongest link observed for customer-related causes, followed by technology-related and service provider-related causes. This pattern implies that when digital interactions falter, customers tend to complain even when precipitating conditions lie partly with users or the surrounding technology, a tendency consistent with attribution bias in high-expectation environments where seamlessness is assumed as the default (Lovelock & Wirtz, 2016).

5.1.2. From customer complaint to justice perceptions

Customer complaint is positively associated with distributive justice, interactional justice, and procedural justice when the complaint is channeled into a recovery episode and handled with some effectiveness. The estimates reflect the net effect of the service provider's response rather than a mechanical consequence of complaining. Post-complaint customers expect appropriate and tangible redress (McColl-Kennedy & Sparks, 2003), empathetic and respectful interpersonal treatment (Vázquez-Casielles et al., 2012), and clear and transparent procedures (Ha & Jang, 2009). Interactional justice shows the largest association, followed by distributive justice and then procedural justice. This pattern

aligns with technology-mediated contact where artificial intelligence and automation can raise efficiency but cannot replace acknowledgment and empathy, which restore a sense of being heard even when technology executes parts of the process (Park & Whang, 2022; Xu et al., 2019). This interpretation, however, does not claim that complaint guarantees fairness, rather, across observed recovery encounters, higher-quality responses are linked to higher justice perceptions, with interpersonal treatment carrying the greatest weight.

5.1.3. From justice perceptions to emotions and outcomes

Perceived justice shapes customer emotion, with interactional justice exerting the strongest influence, followed by distributive justice, and then procedural justice, which is positive yet smaller. These emotions, in turn, raise loyalty, trust, and word-of-mouth, and reduce switching, with the positive path to loyalty being the largest among the three favorable outcomes. This consolidated pattern clarifies mixed findings by showing that in digital contexts, where face-to-face cues are limited, respectful interpersonal treatment and fair outcomes carry greater weight than procedural fairness in restoring emotions and behavior (Crisafulli & Singh, 2017).

5.1.4. Comparison with offline evidence

Findings for online settings diverge from prominent offline meta-analyses that elevate distributive justice and place less emphasis on interactional justice (Orsingher et al., 2010; Ozkan-Tektas & Basgoze, 2017; Roschk & Gelbrich, 2014; Valentini et al., 2020; Van Vaerenbergh et al., 2014; Van Vaerenbergh et al., 2018).

A prominent difference concerns the hierarchy of justice signals. In online recovery, interactional justice is the most influential driver of emotion, distributive justice follows, and procedural justice contributes

least. This ordering reflects mediated contact that strips away face-to-face cues, which raises the marginal value of humanized communication that conveys acknowledgment, empathy, and respect, though compensation and process clarity, nevertheless, still matter yet carry less weight when socioemotional bandwidth is thin.

Another noteworthy difference concerns attribution and controllability. Offline syntheses often link higher complaint rates to failures that sit within service provider control and then tie those complaints to compensation as the most effective lever for satisfaction restoration (Orsingher et al., 2010; Roschk & Gelbrich, 2014; Van Vaerenbergh et al., 2014). The online pattern shifts toward stronger complaint tendencies even when causes appear less controllable by the service provider, as seen in the relatively larger path from customer-related causes and a comparably strong path from technology-related causes to customer complaint, which is consistent with expectations of seamless digital performance and a tendency to hold the service provider accountable for coordination across interfaces and platforms. In this environment, interactional justice addresses the immediate relational breach by restoring a sense of being heard, with distributive justice reinforcing repair where tangible loss is salient, and procedural justice providing necessary, though less decisive, reassurance.

5.2. On variations in online service failure and recovery

Marketing-related (contextual) factors condition effects across the model. Brand prestige changes both expectations and reactions. Complaint propensity after technology-related failure is higher for high-image or prestige brands because customers anchor on elevated standards experience sharper disconfirmation when glitches occur (Maxham and Netemeyer, 2003; Van Doorn & Verhoef, 2008). Offering type also shifts pathways. Evidence has alternated between stronger effects of interactional justice on emotions for product-based services and stronger effects for true services; the consolidated pattern here favors product-based services, where tangible losses and clear redress expectations heighten the value of empathetic, respectful treatment after failure (Chen & Kim, 2019; Harris et al., 2006; Holloway et al., 2005). Purchase frequency further differentiates outcomes. High-frequency customers translate favorable post-recovery emotions into advocacy and retention more readily than low-frequency customers, a pattern that clarifies prior mixed results and points to the importance of managing emotional experience for frequent users (Arsenovic et al., 2023).

Sample-related (methodological) factors similarly help explain several inconsistencies in the online service failure literature. Procedural justice tends to influence emotions more strongly among higher-income customers than among lower-income customers, consistent with evidence on income-linked expectations and entitlement clarity (Ding & Lii, 2016; Keaveney & Parthasarathy, 2001). Country development status also matters. Failures attributable to service provider-related causes prompt more complaints in developed markets, whereas technology-related failures prompt more complaints in developing markets, a pattern aligned with stronger external blame tendencies where institutional reliability is less certain (Rohden and Matos, 2022; Weber & Sparks, 2010). These differences suggest that customers in developed markets evaluate failures against higher baselines for process control and service quality while customers in developing markets display greater external attribution when technology falters (Malhotra et al., 2005; Yan et al., 2021). In practical terms, frontline responses in developed markets should foreground interactional justice through empathetic, culturally sensitive communication while recovery in developing markets should emphasize technology reliability and rapid troubleshooting (Chelminski & Coulter, 2011; Yan et al., 2021).

Combined effects across marketing-related and sample-related moderators also emerge. Emotional responses to fair recovery drive loyalty and word-of-mouth more strongly among higher-income customers interacting with high-prestige brands, reflecting the weight such customers place on symbolic brand attributes and credible performance

under stress (Das et al., 2024; Lim et al., 2025). Strategy therefore benefits from segmenting by income and calibrating recovery to prestige positioning, whereby prestige brands should match elevated expectations with visible interactional care and appropriate redress while non-prestige brands gain by signaling procedural clarity and dependable outcomes, particularly when serving lower-income segments or markets with weaker institutional reliability (Chelminski & Coulter, 2011; Malhotra et al., 2005; Yan et al., 2021).

6. Conclusion

This meta-analysis establishes a coherent failure-to-recovery chain for online services, whereby customer-related, service provider-related, and technology-related causes increase customer complaint (strongest for customer-related), effective handling of such complaint elevates perceptions of distributive, interactional, and procedural justice (interactional most responsive), those justice perceptions lift customer emotion (again led by interactional), and such emotion raises loyalty, trust, and word-of-mouth while lowering switching. Moderator patterns (i.e., brand prestige, offering type, and purchase frequency on the marketing side and age, gender, income, and status of country development on the sample side) account for cross-study discrepancies and signal where recovery should be segmented and tailored. The online pattern departs from offline evidence by placing interactional justice, rather than distributive justice, at the center of emotional repair, which reframes both theory and practice for digital settings. The sections that follow detail the theoretical contributions, managerial implications, and directions for future research.

6.1. Theoretical contributions

This study makes four contributions to service failure research in general and to online service failure in particular.

First and foremost, the meta-analysis herein treats online service failure as a full process (i.e., causes of online service failure to customer complaint to justice perceptions to customer emotion to customer subsequent behavior) within an integrated lens that combines attribution theory and justice theory. Attribution theory (Weiner, 1985) explains why customers frequently hold service providers responsible for failures arising from customer-related, service provider-related, and technology-related causes, a bias that manifests as customer complaint. Justice theory (Greenberg, 1990) then explains how post-complaint expectations of distributive, interactional and procedural justice shape customer emotion and downstream behavior. Alignment between expected and received justice tends to elicit positive emotions and favorable outcomes, whereas misalignment tends to generate negative emotions and switching intentions. A large evidence base (147 studies, 82,901 individuals, 24 countries) yields convergent and precise support for these links and clarifies where effect sizes are strongest within the chain.

Second, the moderation findings extend theory by showing when and for whom complaining tendencies, justice perceptions, and emotions are most affected. Country development status and income shift key paths in ways consistent with established frameworks. In more individualistic, higher-expectation contexts that characterize many developed markets, accountability concerns heighten complaint tendencies when failures are attributed to service providers (Hofstede, 1980; Weber & Sparks, 2010). Where digital infrastructure is thinner and technology hurdles are more salient, technology-related failures more readily trigger complaints and negative emotions, consistent with the unified theory of acceptance and use of technology's (UTAUT) emphasis on effort expectancy and facilitating conditions (Malhotra et al., 2005; Venkatesh et al., 2003; Yan et al., 2021). Offering type and purchase frequency also matter, as interactional justice has a stronger impact on emotions in product-based settings where tangible loss and clear redress expectations heighten the value of empathetic and respectful treatment (Chen & Kim, 2019; Harris et al., 2006; Holloway et al., 2005) while emotions

more readily translate into loyalty and word-of-mouth among frequent users who have clearer reference points for recovery quality (Arsenovic et al., 2023). These patterns open avenues to connect online service failure with cultural dimensions theory (Hofstede, 1980) and with affective events theory (Weiss & Cropanzano, 1996), which predicts stronger emotional responses when customers appraise service encounters as personally significant.

Third, the meta-analytic evidence herein elevates interactional justice as the pivotal driver of customer emotion in online recovery, exceeding the influence of distributive and procedural justice. Customer complaint shows the strongest association with interactional justice and this form of justice shows the strongest association with customer emotion, which then shape their loyalty, switching, trust, and word-of-mouth. This emphasis, in turn, implies boundary conditions for automated recovery. Noteworthy, automated agents built on artificial intelligence improve consistency and increase speed yet often lack empathic signals that support emotional repair and interactional justice, and thus, human agents can supply those signals while well-designed anthropomorphic cues can narrow the gap (Pavone et al., 2022; Shah et al., 2025). Hybrid recovery models (e.g., automated triage plus human engagement during moments that matter) also fit the pattern of effects observed and translate the theoretical primacy of interactional justice into design guidance.

Last but not least, the online context differs meaningfully from offline context of service failure, thus warranting separate theorizing and evidence synthesis. Notably, influential offline meta-analyses tend to place distributive justice at the center of recovery satisfaction and to assign smaller roles to interactional justice (Orsingher et al., 2010; Roschk & Gelbrich, 2014; Van Vaerenbergh et al., 2018; Valentini et al., 2020). The present findings reverse that emphasis for online settings, with interactional justice exerting the strongest effect on customer emotion and distributive justice next, and thus, they uniquely document how customer-related, service provider-related, and technology-related causes differentially heighten customer complaint—an attribution layer not unpacked by prior syntheses (Ozkan-Tektas & Basgoze, 2017; Van Vaerenbergh et al., 2014). Characterizing online service failure as a distinct phenomenon in this way, therefore, advances theory on service failure and recovery in digital environments and sets a clearer agenda for future research.

6.2. Managerial implications

Online service failures continue to disrupt service delivery despite advances in digital channels, which requires deliberate recovery design. The evidence points to several actionable priorities.

First and foremost, interactional justice deserves primacy in online recovery, since empathetic, respectful communication most strongly shapes customer emotion. Automated agents can be equipped to recognize and respond to affect through natural language processing and emotion recognition, supported by service scripts that adapt content and tone to customer state—a direction that balances relational value with operational efficiency and is consistent with service-dominant logic (Vargo & Lusch, 2004, 2008) and social response theory (Nass & Moon, 2000; Reeves & Nass, 1996), which both emphasize responsive, relational exchange.

Second, recovery should be individualized by linking customer profiles, failure causes, and expected justice dimensions within customer relationship management systems that surface guidance to agents in real time. During payment failures, for instance, prompt identification of affected customers and targeted responses that explain steps taken, timelines, and restitution levels can raise perceived procedural clarity and distributive fairness at the moment they matter most.

Third, sensitivity to technology-related failures varies with brand prestige, gender composition, and country development status. In markets with weaker institutional reliability, customers tend to externalize blame when technology falters, which elevates complaints for

technology-related incidents, and such instances, rapid troubleshooting and clear status updates should, therefore, be front-loaded, together with resilient back-end processes. Therefore, prestige brands should anticipate sharper reactions when expectations are violated and pre-commit to visible recovery standards for technology-related incidents while avoidance of stereotypes and gender-sensitive language should be embedded in both automated and human interactions to reduce friction in complaint handling.

Fourth, exposure to service provider-related failures is not uniform across brands and offerings. Tighter margins for error arise for high-prestige brands on technology-related failures and for product-based services where interactional treatment most strongly shapes customer emotion, and thus, high-prestige brands should pre-commit to rapid diagnostics, transparent status updates, and proportionate restitution for technology-related incidents, whereas providers of product-based services should foreground empathetic, respectful interpersonal communication and close the loop with simple procedures and fair outcomes while targeted loyalty benefits can recognize customers who remain engaged through recovery.

Fifth, complaints arising from customer-related failures are more likely when customers are less experienced or engage infrequently, which argues for clear pre-emptive guidance, process simplification, and timely micro-education at points of friction, thus error-prevention design patterns along with instructional nudges and walkthroughs lower the probability of failure attribution spiraling into complaints.

Sixth, the translation from customer complaint to justice perceptions is more elastic under prestige conditions, with the strongest movement observed for interactional justice, therefore, frontline teams should be trained to communicate with empathy and precision while complaint systems should make procedures transparent and outcomes proportionate to loss, using clear timelines and status cues that customers can track without effort.

Seventh, the impact of interactional justice on customer emotion is stronger for prestige brands and varies by offering type and market segment, therefore, personalization at the level of salutation, recall of prior interactions, and explicit acknowledgment of inconvenience, followed by post-resolution follow-ups, strengthens emotional repair and translates into loyalty, trust, and word-of-mouth, particularly for product-based services and in higher-expectation markets.

Last but not least, across these actions, context and segment should guide emphasis, for instance, prestige positioning heightens expectations and penalties for failure, technology-related incidents in developing markets require robust diagnostics and clear communication, and customers who are inexperienced or infrequent benefit most from preventive design and just-in-time guidance, so that interactional care, procedural clarity, and fair outcomes are delivered in concert and at speed.

6.3. Limitations and future research directions

This study enriches the literature on online service failure while carrying several limitations.

First and foremost, the quantitative scope of the meta-analysis excluded qualitative studies and the English-language restriction narrowed coverage to a subset of the global evidence base. Generalizability would improve by incorporating non-English scholarship and qualitative insights that unpack context, process, and meaning.

Second, several moderation effects could not be tested because underlying primary studies did not report sufficient statistics. For instance, interactions of customer-related causes with brand prestige, income level, and status of country development status could not be examined. Harmonization choices during coding, despite high inter-coder agreement, may also leave residual category ambiguity when primary labels differ for similar constructs.

Third, future research should extend the theoretical lens beyond attribution theory and justice theory. For instance, affect control,

commitment-trust, equity, and social exchange perspectives have informed studies of service failure (Lim et al., 2025) and offer complementary predictions about how customers interpret loss, assign responsibility, and re-evaluate service providers across episodes. The next step, in turn, is a meta-analytic model that nests these perspectives and tests where each is most predictive along the chain from causes to customer complaint to justice perceptions to emotions to outcomes.

Fourth, temporal dynamics merit focused attention. Cross-sectional designs dominate the current evidence, yet perceptions after an online service failure evolve as customers receive updates, experience partial remedies, or encounter subsequent transactions. Longitudinal designs (e.g., event-study analyses around incident windows, field experiments that manipulate recovery levers over time, panel surveys) would clarify how justice signals cumulate or decay and when emotions convert into durable advocacy, loyalty, trust, or exit.

Fifth, cultural context remains underexplored. Culture shapes expectations for accountability, compensation, and interpersonal treatment, as dimensions such as individualism, power distance, and uncertainty avoidance (Hofstede, 1980) likely condition the direction and strength of links in the model. The MASEM herein, however, could not include these moderators due to insufficient reporting in the primary literature. Hence, purpose-built primary studies should test them directly and report statistics that enable future syntheses.

Last but not least, measurement and method improvements would strengthen inference. Greater use of behavioral outcomes (e.g., actual switching, repeat purchase, and observed word-of-mouth) alongside self-reports, standardized reporting of reliability and intercorrelations, and consistent operationalization of justice components would reduce unexplained heterogeneity. Mixed-method designs that pair qualitative tracing of recovery episodes with quantitative estimation can further reveal mechanism while preserving comparability, which, in turn, would enable finer-grained moderator tests and help map which recovery levers matter most for which customers and contexts.

CRedit authorship contribution statement

Manish Das: Writing – review & editing, Writing – original draft, Conceptualization, Investigation. **Charles Jebarajakirthy:** Writing – review & editing, Validation, Supervision, Project administration, Conceptualization, Investigation. **Haroon Iqbal Maseeh:** Writing – review & editing, Software, Methodology, Investigation, Formal analysis. **Weng Marc Lim:** Visualization, Supervision, Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Jinal Sameer Shah:** Writing – original draft, Investigation, Data curation.

Declaration of competing interest

Weng Marc Lim has served as an Associate Editor of the *Journal of Business Research* and was excluded from editorial decision-making related to the acceptance of this article for publication in the journal.

Data availability

Data will be made available on request.

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